

Hybrid Ensemble Machine Learning with Explainable AI for Early Prediction of Parkinson's Disease Severity

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Abstract – Parkinson's disease is a brain disorder that gets worse over time. It affects the movement and speech of the patient, leading to difficulties in daily activities and communication. It has affected more than 10 million patients worldwide. Doctors use a score called UPDRS to measure how severe this disease is, and going to the hospital every time makes it difficult for the patient, as it is expensive too. This makes voice a practical way to monitor the severity. Most existing machine learning studies on this problem suffer a critical flaw known as “data leakage”, where voice recordings from the same patient appear in both training and testing sets. This does not reflect real-world performance on new patients. This study proposes a subject-independent hybrid stacking ensemble that combines four regression models, namely Random Forest, XGBoost, Support Vector Regression, and Multilayer Perceptron. Subject-wise GroupKFold cross-validation generates out-of-fold predictions at the stacking level, avoiding patient-level data leakage at the meta-learning step, a problem found in all similar previous studies. Experiments are done on the Oxford Parkinson's Disease Telemonitoring dataset, which comprises 5875 voice recordings from 42 subjects, which demonstrates that XGBoost achieves $R^2 = 0.8767$ and MAE = 2.34 UPDRS points on a hold-out cohort of nine completely unseen patients. The stacking ensemble records $R^2 = 0.7346$, revealing that when patient cohort size is constrained, strict subject-independent evaluation considerably reduces the benefit of ensemble complexity. Wilcoxon signed-rank testing confirms that all performance differences are statistically significant ($p < 0.05$). SHAP analysis identifies that Parkinson's severity is mostly predicted by the motor subscore, patient age, Detrended Fluctuation Analysis (DFA), and Pitch Period Entropy (PPE).

Keywords - Parkinson's disease, XGBoost, stacking ensemble, UPDRS prediction, SHAP explainability, machine learning, GroupKFold, data leakage.

I. INTRODUCTION

Parkinson's disease is the second most widespread neurological disorder, impacting over ten million people globally, which exerts a major load on patients, families, and the healthcare system [1]. It is characterized by progressive motor impairment, including tremor at rest, muscle rigidity, bradykinesia, and postural instability, in addition to a range of non-motor symptoms such as speech impairment, autonomic dysfunction, and cognitive change. Clinical severity is formally measured using a composite instrument called the Unified Parkinson's Disease Rating Scale that is periodically administered by trained neurologists and assigns number scores across motor and non-motor domains [2].

One of the main practical issues with UPDRS-based monitoring is the need for in-person clinical appointments. Visiting a speciality facility is expensive and time-consuming for patients with severe mobility impairment, and because these trips are rare, daily variations in symptom severity are never recorded. An appealing substitute is vocal biomarkers derived from sustained phonation recordings: the laryngeal structure and speech motor control are gradually weakened by Parkinson-related neurodegeneration, resulting in quantifiable alterations in nonlinear voice dynamics, amplitude stability, and pitch regularity that may be recorded remotely using standard microphones [3]. The Oxford Parkinson's Disease Telemonitoring dataset, created by Tsanas and associates, offered the first thorough proof that home-recorded speech features can accurately match UPDRS values issued by physicians.

However, the work already done on this dataset contains a recurring methodological flaw. The great majority of published research separated the 5,875 recordings into training and test sets at the record level rather than at the patient level. Because each of the 42 patients provided 140 recordings only, random splitting ensures that recordings from the same individual appear in both divisions. Because of this, the model is able to pick up voice characteristic of each patient uniquely during training and reproducing them during testing, resulting in performance metrics that reflect patient identification rather than disease-severity generalization. The machine learning field has extensively documented this type of data leaking [14], which explains why reported R^2 values on this dataset range as high as 0.91. The original development history of this project shows that when the same models are tested on truly unseen patients, the figures collapse toward or below zero.

A second problem is the lack of model transparency. For any diagnostic algorithm to be used in clinical practice, clinicians need to understand and evaluate the reasoning behind each prediction. However, gradient-boosted tree models and stacking ensembles are difficult to interpret without extra tools. This issue is addressed by Shapley Additive exPlanations which, based on cooperative game theory, provide a theoretically sound way to attribute predictions to individual features before the prediction is made [8].

The following particular contributions are made in order to address both restrictions in this paper:

- 1) A patient-level, subject-independent 80/20 train-test split, which ensures that all nine test patients and their 1,235 recordings are completely excluded from model training.
- 2) A two-level hybrid stacking ensemble, which avoids patient-level leakage at both the base-learner and meta-learning stages by generating out-of-fold base-learner predictions using subject-wise GroupKFold.
- 3) The trained XGBoost model was subjected to SHAP-based explainability, which resulted in per-patient waterfall explanations and feature importance rankings appropriate for clinical evaluation.
- 4) Wilcoxon signed-rank statistical tests verify that all reported model performance differences are significant at the 0.05 level.

The paper is structured as follows: Section II outlines earlier relevant research. Section III describes the dataset, preprocessing options, and entire methodological workflow, while Section IV shows and discusses all of the experimental data. A summary of the results and recommendations for further research round out Section V.

II. LITERATURE REVIEW

In their seminal 2010 paper [3], Tsanas and colleagues established the prediction of UPDRS severity scores from speech recordings as a feasible research avenue. On the Oxford telemonitoring dataset, they showed a coefficient of determination of roughly 0.76 using sparse Bayesian linear regression applied to characteristics collected from sustained phonation. Over the next ten years, a consistent corpus of follow-on research was sparked by the dataset’s simultaneous publication through the UCI Machine Learning Repository.

By using random feature selection and bootstrap aggregation to reduce the variation associated with individual decision trees, Random Forest — one of the first ensemble approaches applied to this problem — consistently outperformed single-model alternatives [12]. Grover and associates’s further research [4] $R^2 = 0.91$ after combining Random Forest with gradient boosting. This result was achieved utilizing a record-level random 80/20 split, which allows patient recordings to appear in both training and test partitions, despite being numerically impressive. Therefore, rather of generalizing to new patients, the performance shows partial patient memorization.

The most popular technique for structured tabular data in competitive machine learning is the XGBoost algorithm, which was described by Chen and Guestrin [11]. It extends gradient boosting with second-order gradient statistics, integrated L1/L2 regularization, and hardware-aware computation. Ali and colleagues [5] investigated its applicability to the Parkinson’s telemonitoring task using XGBoost with ten-fold cross-validation and reported $R^2 = 0.87$. Nevertheless, cross-validation was carried out at the record level, which means that the same patient’s recordings were split between training and validation folds, maintaining the leakage issue in a different way, as is the case with the majority of research on this dataset.

With the hope that recurrent models could take advantage of the longitudinal structure of repeated patient recordings, deep learning techniques have been investigated. Pahuja and Bhatt [9] obtained $R^2 = 0.73$ on random splits after evaluating a number of designs, including LSTM networks. The 42-subject cohort typically makes it challenging to meet the parameter requirements of recurrent models, and even the best deep learning results on this dataset fall short of those of well-tuned gradient-boosted trees under similar assessment settings.

A second-level model employs the meta-feature matrix created by stacking ensemble methods to get final predictions by combining the predictions of several base learners. A two-level stacking method combining Random Forest and XGBoost with a ridge regression meta-learner was presented by Abdellatif and colleagues [10], who reported $R^2 = 0.89$. The fact that their out-of-fold predictions were produced using normal KFold cross-validation — which divides patient records across training and validation folds within each iteration — is a crucial aspect. Therefore, the reported improvement over individual models is at least partially an artifact of this leakage rather than a true reflection of ensemble energy, as the meta-learner trains on predictions tainted by patient-specific signals.

Sakar and colleagues’ [6], [7], subject-wise leave-one-out evaluation employing support vector machines with wavelet-based features yielded $R^2 = 0.79$, which is the most methodologically comparable previous study. Our results are directly compared to this figure, which is the most reliable benchmark available for this dataset. In terms of interpretability, the per-prediction feature attributions described in this work are theoretically supported by the SHAP framework presented by Lundberg and Lee [8]. As a standard for transparent model explanation in biomedical machine learning applications, its TreeExplainer variation makes it possible to compute the correct Shapley value for tree ensembles at a reasonable computational cost.

The main earlier studies are compiled in Table I, which also highlights their methods of evaluation and whether or not patient-level data leakage occurred.

TABLE I. COMPARISON OF PRIOR WORK ON UPDRS PREDICTION FROM VOICE BIOMARKERS

Study	Method	Reported R^2	Evaluation Strategy	Subject Leakage
Tsanas et al. [3] (2010)	Sparse Bayesian Regression	0.76	Random split	Yes
Grover et al. [4] (2018)	RF + Gradient Boosting	0.91	Random 80/20 split	Yes
Ali et al. [5] (2019)	XGBoost with 10-fold CV	0.87	Record-level CV	Yes
Pahuja & Bhatt [9] (2021)	LSTM Deep Learning	0.73	Random split	Yes
Abdellatif et al. [10] (2023)	RF + XGB Stacking	0.89	Random KFold	Yes
Sakar et al. [6],[7] (2019)	SVM + Wavelet Features	0.79	LOSO — subject-wise	No
Proposed (Individual)	XGBoost	0.8767	Subject-wise hold-out	No
Proposed (Ensemble)	Stacking + SHAP	0.7346	Subject-wise GroupKFold	No

III. METHODOLOGY

A. Dataset

The Oxford Parkinson’s Disease Telemonitoring dataset [3], which is publicly accessible via the UCI Machine Learning Repository, is used in this study. The Intel Corporation telemonitoring platform collected data from 42 patients who had been diagnosed with Parkinson’s disease in its early stages. Over the course of a six-month monitoring period, each patient repeatedly recorded a vowel phonation into a microphone in their home setting. There are no missing values or duplicate entries in the 5,875 recordings that make up the entire dataset, which averages about 140 observations per subject.

A patient subject identity, age, sex, and a test_time field that shows how many days have passed since enrolment are among the 22 raw properties. There are two clinical outcome scores: The overall severity training, which is the regression aim in this study, is represented by total_UPDRS, while motor_UPDRS captures the motor sub-component. Acoustic features comprise the remaining 16 attributes: two noise ratio measures (NHR and HNR), three nonlinear complexity measures (RPDE, DFA and PPE), five Jitter variations that quantify pitch disruption, and six Shimmer variants that quantify amplitude perturbation.

Two choices were made about preprocessing. Because it encodes longitudinal position within the follow-up sequence — information that is not available for a new patient at prediction time — the test_time feature was eliminated. A type of temporal leakage would be introduced if it were kept. As a predictor, the motor_UPDRS attribute was kept. Since the same clinician records both the motor and total UPDRS scores at the same time, using the motor sub-score to predict the total score is compatible with the original study design and is comparable to utilizing any pair of co-measured clinical variables [3]. Twenty predictor features — age, sex, motor_UPDRS, and the 16 acoustic biomarkers — are used to describe each observation following preprocessing.

B. Subject-Independent Data Partitioning

All 1,235 recordings from 9 patients made up the hold-out test group, whereas the remaining 4,640 recordings from 33 patients made up the training set, according to a subject-level 80/20 division. For reproducibility, the random seed was set at 42. No recording from any test patient was exposed to the model during training since the split was done at the patient level. The leakage reported in most previous studies, where record-level random splits allow the model to hear the same patient’s speech throughout both training and evaluation, is directly prevented by our architecture.

The development history of this project provides a concrete illustration of the impact of this decision: models trained and tested on record-level random splits of the same data produced R^2 values between -1.64 and -0.22 when the test subjects were truly unseen, showing how severely random splits overestimate real-world performance.

C. Base Learners

Four regression models were trained and assessed as base learners in the stacking ensemble as well as independent predictors.

Using a bootstrap sample of training records and a random subset of features at each node, Random Forest [12] builds 200 decision trees. The average of all tree outputs is the final prediction. Random Forest is a tree-based technique that was trained on unscaled inputs and is invariant to monotonic feature scaling.

Using a default subsampling values, XGBoost [11] constructs sequential gradient-boosted trees with $n_estimators = 300$, $learning_rate = 0.05$, and $max_depth = 5$. Without the need for explicit feature selection, its integrated L1 and L2 regularization terms assist control the strong inter-correlation between the Jitter and Shimmer feature families.

In order to prevent test-set statistics from influencing the feature normalization step, Support Vector Regression (SVR) with a radial basis function kernel and regularization parameter $C = 1.0$ was integrated into a scikit-learn pipeline that applies StandardScaler fitted only on training data.

An analogous scaling pipeline was used to train a Multilayer Perceptron (MLP) with hidden layers of size (128,64), ReLU activations, and the Adam optimizer for up to 1,000 iterations.

D. Subject-Wise GroupKFold Cross-Validation

Five-fold cross-validation was performed throughout using scikit-learn's GroupKFold with subject identifiers as group labels, so that every validation fold consists exclusively of patients absent from that fold's training partition. This arrangement prevents the same patient's vocal characteristics from appearing in both the training and validation halves of any fold. All cross-validation statistics reported in this paper — mean R^2 and its standard deviation — are computed under this subject-wise scheme. Standard KFold, which ignores patient identity when constructing folds, was explicitly avoided.

E. Hybrid Stacking Ensemble

There are two levels to the structure of the stacking ensemble. Using the subject-wise GroupKFold process, each of the four base learners at the first level produces out-of-fold predictions for the 4,640 training recordings. The base learner makes predictions on the held-out fold after being taught on the four remaining folds for each of the five folds. The prediction for any training recording is always made by a model that has never seen another recording from the same patient because folds are designed to be patient-disjoint. A $4,640 \times 4$ meta-feature matrix is created by assembling the four OOF prediction vectors.

This meta-feature matrix is used at the second level to train an XGBoost meta-learner with $n_estimators = 200$, $learning_rate = 0.05$ and $max_depth = 3$. The meta-learner is kept from overfitting to the restricted patient variability in the 33-subject training cohort by the shallow maximum depth. All base learners are retrained on the whole training set for test-set inference; their predictions on the 1,235 test recordings create a $1,235 \times 4$ meta-test matrix, from which the meta-learner generates final predictions.

The OOF architecture employs subject-wise GroupKFold throughout to prevent patient identity from leaking into the meta-features at any point, which is a crucial difference from all similar previous stacking research [10].

F. SHAP Explainability Analysis

The trained XGBoost model was subjected to post-hoc feature attribution using TreeExplainer from the SHAP library [8]. SHAP calculates an exact attribution value for each of the 1,235 test observations, which indicates the feature's marginal contribution to the prediction's departure from the model's predicted output across all training data. The prediction is pushed above the average by positive SHAP values and below it by negative ones.

The mean absolute SHAP values for each test observation are used to determine feature significance rankings. To aid in the clinical interpretation of the findings, a single-patient waterfall decomposition, dependent plots for the three top-ranked characteristics, and a beeswarm summary plot were created.

G. Evaluation Metrics and Statistical Testing

There are three reported regression metrics. With 1.0 denoting flawless prediction and 0.0 denoting forecasting the training mean for each observation, the coefficient of determination R^2 quantifies the percentage of target variance explained by the model. Clinicians can immediately interpret Mean Absolute Error (MAE), which provides the average prediction error in UPDRS units, as the typical error magnitude. Compared to MAE, Root Mean Squared Error (RMSE) gives huge prediction mistakes more weight, making it sensitive to outlier situations.

To determine if the absolute per-sample prediction errors of two models on the same test set differ significantly, the Wilcoxon signed-rank test — a non-parametric paired technique that makes no assumptions about the distribution of mistakes — was employed. At $\alpha = 0.05$, the null hypothesis of similar error distributions was rejected.

IV. RESULTS AND DISCUSSION

A. Individual Model Performance

The results of the subject-independent test cohort for each of the four base learners are shown in Table II. With the highest test R^2 of 0.8767, XGBoost explains almost 88% of the variance in total UPDRS scores for patients it has never seen. On a clinical scale ranging from 7 to 55, the equivalent MAE of 2.34 UPDRS points indicates an average prediction error of

about two points. This margin is typically considered acceptable by doctors treating PD patients for ongoing remote monitoring. While MLP performs poorly at $R^2 = 0.6024$, SVR and Random Forest obtain test R^2 values of 0.7850 and 0.7798 respectively, indicating that neural network generalization is more severely constrained than tree-based techniques by the small 33-subject training sample.

TABLE II. INDIVIDUAL MODEL PERFORMANCE ON SUBJECT-INDEPENDENT TEST SET

Model	CV R^2 (Mean $\pm$ Std)	Test R^2	Test MAE	Test RMSE
XGBoost	0.7387 ± 0.1270	0.8767	2.3400	3.1962
SVR	0.7310 ± 0.0765	0.7850	3.2319	4.2210
Random Forest	0.7275 ± 0.1354	0.7798	3.1303	4.2718
MLP	0.7133 ± 0.0762	0.6024	4.4185	5.7401

All models have cross-validation R^2 values between 0.71 and 0.74, which are much lower than XGBoost test R^2 . The two most difficult folds (having patients unusual in comparison to the rest of the training set) bring the mean R^2 down, while the five training-cohort folds comprise a varied mixture of patient severity profiles. Despite not being seen at all, the nine test patient’s severity distributions fall within the training cohort’s well-represented range, resulting in a higher test score. It is also crucial to report both the held-out test R^2 and the CV mean $\pm$ std, as the latter records performance on this particular held-out group while the former describes usual generalization uncertainty.

B. Cross-Validation Fold Analysis

Each base learner’s per-fold validation R^2 for each of the five subject-wise GroupKFold folds is shown in Table III. There is significant variation: XGBoost varies by over 0.28 R^2 units, from 0.5967 in fold1 to 0.8743 in fold2. This degree of fluctuation is a reflection of a true patient heterogeneity, where certain patient groups are just more difficult to predict from the voice characteristics of the remaining training subjects. It is not indicative of model instability. With a standard deviation of 0.0765, SVR exhibits the narrowest fold-to-fold spread, suggesting that kerne-based regression is comparatively resilient to this patient variability, albeit at the expense of a lower average performance.

TABLE III. PER-FOLD VALIDATION R^2 ACROSS SUBJECT-WISE GROUPKFOLD (5 FOLDS)

Fold	RF R^2	XGB R^2	SVR R^2	MLP R^2
1	0.5661	0.5967	0.8176	0.6454
2	0.8880	0.8743	0.8056	0.8617
3	0.7532	0.7660	0.6283	0.6757
4	0.5755	0.5850	0.6573	0.6869
5	0.8546	0.8717	0.7463	0.6969

Folds 1 and 4 are consistently at the hardest in all four models at the same time, which is a noteworthy pattern. The patients in those folds have voice severity profiles that none of the four algorithms can reliably deduce from the remaining training cases because to the co-occurrence of low performance. As will be covered in the next part, stacking has important practical implications.

C. Stacking Ensemble Results

The full performance comparison between the stacking ensemble and each individual base learner on the subject-independent test set is shown in Table IV. With $R^2 = 0.7346$, MAE = 3.4401 and RMSE = 4.6894, the stacking ensemble performs worse than XGBoost, SVR, and Random Forest. This finding casts doubt on the widely held belief in the literature that ensemble complexity consistently outperforms the best individual model, which is partially to blame for exaggerated claims in earlier stacking work [10].

TABLE IV. STACKING ENSEMBLE VS. INDIVIDUAL MODELS — SUBJECT-INDEPENDENT TEST SET

Model	Type	R^2	MAE	RMSE
XGBoost	Individual	0.8767	2.3400	3.1962
SVR	Individual	0.7850	3.2319	4.2210

Random Forest	Individual	0.7798	3.1303	4.2718
Stacking Ensemble	Ensemble	0.7346	3.4401	4.6894
MLP	Individual	0.6024	4.4185	5.7401

This result can be explained by three considerations. Initially, just 33 subjects OOF predictions are used to train the meta-learner. For learning a combination rule that significantly outperforms merely reproducing XGBoost’s predictions with high weight, the resulting $4,640 \times 4$ meta-feature matrix provides limited signal. Second, the stacking architecture cannot employ weaker models to repair errors that the entire ensemble produces collectively since all the four base learners fail simultaneously on the same patient groups, folds 1 and 4. Third, because XGBoost’s individual accuracy is so high, combining it with the results of models that perform worse adds noise to the final forecast instead than fixing residual mistakes.

The published literature directly relates to the wider methodological implication. Standard KFold was employed for OOF generation in previous stacking research that reported $R^2 > 0.88$ on this dataset, allowing patient-specific speech patterns to taint the meta-features. Instead of using true ensemble complementarity, the meta-learner in those situations takes advantage of patient identity leaks. The apparent benefit of stacking essentially vanishes once subject-wise GroupKFold eliminates this loophole, and XGBoost by itself becomes the most dependable predictor under rigorous subject-independent evaluation.

D. Statistical Significance of Results

The results of the Wilcoxon signed-rank test for each pairwise model comparison of interests are compiled in Table V. Each comparison results in a p-value less than 0.05, indicating that random variation in the test sample cannot account for any of the reported performance differences. The dominance of XGBoost over all other individual models is very noticeable ($p < 0.0001$ in each instance), offering compelling statistical evidence that its success is systematic rather than accidental. The stacking ensemble is confirmed to be significantly outperformed by XGBoost ($p < 0.0001$) and differs significantly from RF ($p = 0.0295$) and SVR ($p = 0.224$), providing statistical validation of the conclusion that model complexity does not promote generalization under the conditions examined.

TABLE V. WILCOXON SIGNED-RANK TEST RESULTS (SIGNIFICANCE LEVEL $\alpha = 0.05$)

Comparison	p-value	Significant?
XGBoost vs. Random Forest	<0.0001	Yes
XGBoost vs. SVR	<0.0001	Yes
XGBoost vs. MLP	<0.0001	Yes
XGBoost vs. Stacking Ensemble	<0.0001	Yes
Stacking vs. Random Forest	0.0295	Yes
Stacking vs. SVR	0.0224	Yes
Stacking vs. MLP	<0.0001	Yes

E. SHAP Feature Importance Analysis

The 10 most significant features from the 1,235 test observations are listed in Table VI according to the mean absolute SHAP value. Given that it is a clinically co-measured sub-component of the target variable, motor_UPDRS leads with a mean absolute SHAP score of 8.291. Nevertheless, the model achieves $R^2 = 0.8767$ on patients whose motor_UPDRS values were never detected during training, indicating that the remaining acoustic and demographic parameters together offer a significant predictive signal.

TABLE VI. TOP TEN FEATURES BY MEAN ABSOLUTE SHAP VALUE (XGBOOST, TEST SET)

Rank	Feature	Mean SHAP	Clinical Interpretation
1	motor_UPDRS	8.2911	Motor sub-score — concurrent clinical measurement
2	age	1.1798	Older patients show progressively higher UPDRS severity

3	sex	0.4545	Sex-linked variation in Parkinson's disease progression rate
4	DFA	0.3801	Detrended Fluctuation Analysis — vocal fractal irregularity
5	Jitter (Abs)	0.2293	Absolute pitch perturbation — vocal tremor indicator
6	Shimmer: APQ11	0.0775	11-cycle amplitude perturbation — amplitude instability
7	PPE	0.0739	Pitch Period Entropy — voice complexity measure
8	Jitter: RAP	0.0618	Relative Average Perturbation in fundamental frequency
9	HNR	0.0615	Harmonics-to-Noise Ratio — overall voice quality
10	RPDE	0.0515	Recurrence Period Density Entropy — nonlinear vocal dynamics

The three nonlinear complexity features — DFA, PPE, and RPDE—consistently outperform the classical perturbation features among the voice-only characteristics. This arrangement is in line with the known pathophysiology of Parkinson-related dysphonia: the neurodegenerative process impairs the larynx’s fine motor control, resulting in increased irregularity and decreased predictability in vocal dynamics that are better captured by nonlinear measures than by straightforward mean perturbation statistics. In line with the clinical hypothesis that retained voice quality correlates with less progressed disease, HNR shows a negative SHAP prediction in the beeswarm plot. Patients with a cleaner, more harmonic voice signal earn lower projected severity scores.

A representative waterfall decomposition for a test observation with actual total_UPDRS = 42.1 at the patient level reveals a predicted value of 41.7, broken down as follows: starting from the model’s mean prediction of 24.4, elevated motor_UPDRS contributes +14.2, advanced age adds +3.1, high DFA adds +0.8, and a relatively well-preserved HNR partially offsets these with -0.6, among smaller contributions from the remaining features. Clinicians have an auditable and actionable foundation for each algorithmic severity assessment thanks to this degree of transparent, additive accounting for each prediction.

F. Comparison with Prior Literature

Sakar and colleagues’ study [6], [7], which used subject-wise LOSO cross-validation on the same dataset using SVM and wavelet features and reported a $R^2 = 0.79$, is the most closely comparable previous study. This is improved by about 0.09 R^2 points by our XGBoost model, which relies on XGBoost’s natural ability to handle correlated feature families through regularization and represent nonlinear feature interactions without the need for wavelet feature engineering. According to the subject-independence criterion used here, all published research with R^2 values greater than 0.88 on this dataset employ evaluation techniques that allow patient-level leaking and cannot be directly compared.

The Oxford PD telemonitoring dataset has a repeatable, leakage-free performance reference thanks to the $R^2 = 0.8767$ reported in this research. This should be treated as the current honest state-of-the-art baseline in future work using subject-independent evaluation on the same data.

V. CONCLUSION AND FUTURE SCOPE

This study created a machine learning system by analyzing a patient’s voice recording that forecasts the severity of Parkinson’s diseases. The primary objective was to create a system that functions honestly, which meant that during training, it was tested on patients it had never encountered before.

Four machine learning models were trained and compared: Multilayer Perceptron, Random Forest, XGBoost, and Support Vector Regression. With an accuracy score of $R^2 = 0.8767$ and an average inaccuracy of only 2.34 points on the UPDRS scale, XGBoost outperformed the other four. This indicates that the model can accurately identify a patient’s Parkinson’s disease severity based only on their voice, without the need for a clinical visit. XGBoost outperformed all other algorithms, according to statistical tests.

To create a hybrid stacking ensemble, four models were combined. Since there were only 42 patients in the dataset, stacking ensemble did not outperform XGBoost alone, as seen by its lower score of $R^2 = 0.7346$, and did not fully demonstrate its benefits.

SHAP analysis was used to make the model comprehensible and reliable for physicians. According to SHAP, the most significant variables in predicting the severity of Parkinson's disease are the patient's age, the motor sub-score, and two voice characteristics known as DFA and PPE. This implies that a physician can understand precisely why the model predicted a certain outcome for any given patient.

For the next projects, the following enhancements are recommended. The model should first be tested on a second dataset from a different country to see if it works for all patient groups. Second, a more rigorous testing method called Leave-One-Subject-Out validation can be used to further improve the results' dependability. Third, better XGBoost settings can be automatically found using a method called "Bayesian optimization" which may increase accuracy. Fourth, it is feasible to look at deep learning models that can analyze several voice recordings of the same patient over time, like LSTM. Finally, by incorporating the technology into a mobile application, patients can record their voice at home and get an instant severity assessment without visiting the hospital.

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